

Furkan Kaya

Lead LLM Engineer | AI Consultant

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[LinkedIn](#) · [GitHub](#) · [Hugging Face](#) · [Scholar](#) · [Website](#) · [Portfolio](#)

LLM engineer with 5+ years in production LLM and NLU systems. The work is always the same: go into someone else's environment, learn how it actually runs, and ship something that survives production. I build LLM features that non-technical teams use every day, and I own the prompts, the fallback logic, the observability, and the human review loop. Core work: production LLM services (FastAPI, versioned prompts, structured JSON output, precision/recall evaluation), agentic AI (MCP, tool-calling), RAG pipelines, and open-weight model deployment (vLLM, LoRA/QLoRA). I now work as an independent AI consultant, end to end from knowledge base and RAG setup to fine-tuning, on-premise deployment, and agentic architecture.

Experience

Independent AI Consultant

Entegra Bilişim (Client) · Jul 2026 – Present

- **Agentic Support System:** Designed an agentic system that takes actions inside the user session. It covers two tracks: knowledge-base Q&A and troubleshooting actions (license, token, and version checks).
- **Log Infrastructure:** Led the log analysis with the engineering team. The existing logs were not usable for a model, so I designed a new structured event-log schema.
- **Support Intent Analysis:** Building an intent-classification pipeline for voice and chat support tickets. It triages each ticket and routes it to the correct team.

askvela.ai — Co-Founder & AI Architect

Jan 2025 – Sep 2025

- **Agentic AI Platform:** Built an AI agent product for the maritime industry. More than 10 marinas used it in daily operations.
- **Tool Use & MCP:** Designed the tool-calling layer and the Model Context Protocol integration. The agents call external APIs, read databases, and run marine workflows without a human step.
- **System Architecture:** Built the backend as microservices in Node.js and Python. It handles concurrent agent sessions and real-time data.

Usight Tech — Lead LLM Engineer

Dec 2022 – Mar 2026

- **Production LLM Service:** Built a FastAPI service around LLM APIs. Ops teams used it every day across 20+ e-commerce marketplaces (Trendyol, Hepsiburada, Amazon). Prompt templates are versioned and editable without a code change. Pydantic schemas validate every JSON output. Failed calls retry with exponential backoff.
- **Observability & Fallback:** Logged every LLM call: input hash, model version, output, latency, and token count. The logs drive quality monitoring and anomaly detection. When the primary LLM API failed, the service moved to a fallback model instead of stopping.
- **Marketplace Data Transformation:** Designed an NLU pipeline that standardizes product data across marketplaces. It corrects schema mismatches and cut manual operations effort by about 10 times. Bulk listing now runs without a human step.
- **Multimodal Data Enrichment:** Used Vision-Language Models to read attributes from product images. The extra attributes improved attribute matching accuracy and catalog quality.
- **Distillation & Fine-Tuning:** Distilled reasoning behavior from large models into small local models. I generated Chain-of-Thought training data and fine-tuned with Unsloth (LoRA/QLoRA). Quantization made vLLM inference fast enough to run open-weight models on company servers.
- **Quality Evaluation (LLM-as-a-Judge):** Built an automated scoring and error analysis workflow. I ran precision/recall checks against human reviewer samples. Each failure analysis drove the next prompt and dataset change.
- **RAG Systems:** Built end-to-end RAG systems on ChromaDB. Re-ranking improved context relevance and downstream decision quality.

Istinye University AI Laboratory — AI Researcher

Oct 2019 – Dec 2022

- **Bias in Generative AI:** Researched algorithmic bias in generative models. Built APIs that detect and reduce bias in image generation. Published as a book chapter with CRC Press.
- **Computer Vision & Data Annotation:** Built a defect detection system for a factory line (damaged eggs) and a car damage segmentation model. Automated the data labeling workflow.
- **Accessible AI (NLP/Vision):** Built a sign detection and translation system on Google Street View data for visually impaired users.
- Gave technical advice in project meetings and helped the team set direction.
- Competed with the AI Lab team and won 3 awards.

Skills

- **AI & NLP:** LLMs (fine-tuning, distillation, CoT), Prompt Engineering (zero-shot, few-shot), Structured Outputs (JSON schema, Pydantic), VLMs, RAG, Agents (MCP, tool-calling), LLM-as-a-Judge, LLM Evaluation (precision/recall, human-AI agreement), Token Budgeting, Context Window Management.
- **LLM APIs & Models:** Claude, GPT, Gemini, open-weight models (Llama, Qwen, Gemma), Hugging Face Transformers, Unsloth, vLLM.
- **Backend & DevOps:** Python (async, retry/backoff), FastAPI, Docker, REST APIs, Git, Linux, Node.js, SQL (PostgreSQL, MySQL).
- **Data Science:** Pandas, NumPy, EDA, data cleaning, vector databases (ChromaDB), labeling pipelines.
- **Game Dev & AR/VR:** Unity, AR/VR development.
- **Tools:** Claude Code, Hermes.
- **Languages:** Turkish (Native), English (C1).

Education

Erasmus Exchange Program, Mar–Sep 2024 *Koblenz University of Applied Sciences, Germany*

B.Sc. Management Information Systems (English), 2022–2025 *Istinye University, Türkiye* GPA: 3.84/4.00. Full scholarship. Top student. Active in AI Lab projects.

A.A.S. Computer Programming, 2019–2021 *Istinye University, Türkiye* GPA: 3.96/4.00. Full scholarship. Top student.

Workshops & Consulting

- **Nat Yazılım** — Corporate AI training and consulting, from AI fundamentals to in-house project architecture (2025–2026)
- **Akbank People & Culture** — "Applied Generative AI in Human Resources" workshop (Nov 2024) — [LinkedIn Post](#)
- **EU Delegation to Türkiye** — Generative AI workshop for the diplomatic team, in English, Ankara (2024)
- **Gedik University** — Invited instructor, "New Generation AIs in Game Development" workshop (2025) — [LinkedIn Post](#)
- **Usight Tech** — Technical training on LLM fine-tuning for junior engineers and interns (2023–2025)

Publications

- **Ex Machine, Next Machine, and Still Biased Machine** — In *Artificial Intelligence* (pp. 208–234), CRC Press (2024). [pdf](#)
- **Artificial Intelligence and Disabled Individuals: Data-Based Problems and Prejudice** — *Engelsiz Bilişim* (2021). [pdf](#)

Awards & Honors

- **1st Place**, Google DSC & Beşiktaş Municipality Hackathon (2022) — Compost App (Flutter, waste management).
- **2nd Place**, Intertech Metaverse Hackathon (2022) — DenizBank VR/Metaverse Branch PoC.
- **Top 3**, Google Turkey Solution Challenge (2020) — Plasma Donation Mobile App. [GitHub](#)
- **Winner**, Barrier-Free Informatics Awards (2021).
- **International Representation**, FRC U.S.A. Robotics Competition — New York (2017).